BSc Data Science · Semester 6 · Design Thinking & Entrepreneurial Mindset

Design Thinking & Entrepreneurial Mindset

Case Study Project

PROBLEM TOPIC

Women Safety & Harassment in Mumbai

Proposed Solution: SafeWalk Mumbai — A Community Safety App

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Course	TYBSc Data Science
Subject	Design Thinking & Entrepreneurial Mindset
Submission Date	29 March 2026
Total Marks	50

1 Problem Identification

Overview

Women in Mumbai face a persistent and deeply troubling problem: the fear of harassment and lack of safety while commuting late at night or walking through unfamiliar areas. Despite Mumbai's reputation as a relatively safe metro, street harassment, stalking, unsafe public transport, and poorly lit lanes remain everyday realities for women across all age groups and social backgrounds.

Who Faces This Problem?

Who	Details
Primary Users	Women aged 16–45 who commute, study, or work in Mumbai
Students	College girls traveling by local train / auto late at night
Working Women	IT professionals, nurses, retail staff with irregular shift hours
Domestic Workers	Women walking through isolated lanes early morning or evening

When & Where It Occurs

The problem is most acute between **9 PM and midnight** near local railway stations (Dadar, Andheri, Bandra, Ghatkopar), in auto-rickshaw lanes, dark by-lanes in chawl areas, and bus stops with no CCTV coverage. It also occurs during early morning hours (5–7 AM) when domestic workers travel. Festival seasons and weekends see a spike in incidents.

Why Is This Important?

> According to NCRB data, Mumbai records hundreds of harassment cases annually — and experts estimate that over **70% of incidents go unreported** because women fear social stigma, distrust the reporting process, or simply do not know how to get quick help. This problem restricts women's mobility, economic participation, and mental well-being. A technology-enabled community solution could change this reality.

2 User Understanding (Empathy)

To deeply understand the problem, I personally spoke with **three real users** from my college and neighbourhood in Mumbai. Below are their individual profiles and insights.

Priya Sharma · 22 yrs, B.Com Student, Andheri (W)

Problem faced	Travels by local train from Andheri to Goregaon every morning for lectures at Patkar Varde College. Frequently harassed at the Goregaon station exit and in the dark lane leading to the college.
Main frustration	Nobody intervenes when she faces eve-teasing. She feels completely alone and does not know who to call apart from the police (which takes too long).
Current solution	Shares location with her mother on WhatsApp and texts her every 10 minutes. This causes anxiety for both.

Fatima Khan · 35 yrs, Night Shift Nurse, Sion Hospital

Problem faced	Gets off work at 11:30 PM and has to take an auto alone. Auto drivers often take unknown routes; she cannot tell if it is intentional.
Main frustration	The uncertainty about the auto route is terrifying. She once had to call her husband to follow the auto on his bike. Not sustainable.
Current solution	Uses Google Maps to share live location and keeps her phone on call with husband for the entire journey.

Rekha Kamble · 38 yrs, BMC Sanitation Worker (Sweeper), Dharavi

Problem faced	Starts her garbage collection shift at 5:30 AM and walks 1.5 km through dark, isolated lanes in Dharavi before dawn.
Main frustration	She has a smartphone but no data plan and cannot afford to stay on a call. She has been verbally abused twice but never reported it as she feared losing her job if she was late.
Current solution	Carries a small whistle given by her supervisor and tries to walk with a female colleague when possible. Has no other safety mechanism in place.

Key Patterns & Insights

#	Insight	Relevance
1	Priya and Fatima rely on manual WhatsApp location sharing which is reactive; Rekha has no safety tool at all due to no data plan — showing the solution must work for all income levels.	Need for a low-data or SMS-based safety alert system
2	None of the three trust the police helpline for quick response to low-grade harassment.	Need for community-based immediate help network
3	The core fear across all three is having no nearby trusted person.	Solution must connect women to nearby helpers instantly

3 Problem Definition

How Might We Statement

"How might we help women in Mumbai feel immediately safe and connected to nearby trusted people while commuting alone at night, so that they can travel confidently without fear of harassment?"

Why This Problem Matters

Women's safety is not just a personal concern — it is an economic and social issue. When women feel unsafe, they avoid late-night work, night classes, and social activities. This directly restricts their career growth, education, and freedom. Mumbai, as a global financial hub, cannot afford to keep half its workforce living in fear. A scalable, tech-enabled safety solution has both social impact and significant business potential.

The Insight That Led Here

Key insight from all three users: they do not need more ways to call the police. They need a way to instantly signal distress and get help from **verified, nearby people** — friends, volunteers, or fellow commuters — within a 500-metre radius. The solution must be **fast, discreet, and community-powered.**

4 Ideation — Generating Solutions

Using brainstorming and divergent thinking techniques, I generated the following five solution ideas before selecting the most impactful one:

#	Idea	How It Solves the Problem
1	**SafeWalk App — Community SOS Network**	Mobile app where pressing a panic button sends instant alerts to verified volunteers within 500m, shares live GPS, and auto-dials emergency contacts.
2	Safe Zones Map	Crowdsourced map highlighting safe/unsafe streets in real-time, updated by community reports.
3	Women-Only Auto/Cab Pool	Platform connecting women commuters with women auto/cab drivers for verified safe rides.
4	AI-Powered Street Lighting Request System	Civic-tech app to report and track unsafe dark lanes to BMC with automated follow-up.
5	Buddy Escort Volunteer Network	WhatsApp/app-based network where registered volunteers accompany women for short distances near railway stations.

Selected Idea: #1 — SafeWalk Mumbai App

Why this idea? Unlike the others, SafeWalk directly addresses the core user insight: women need instant community help, not bureaucratic reporting. It works in real-time, is low-cost to build, requires no hardware, and operates entirely on existing smartphones. The community volunteer model creates social trust and local accountability. It is the most practical and impactful solution for Mumbai's dense, connected urban environment.

5 Proposed Solution

SafeWalk Mumbai — How It Works

SafeWalk Mumbai is a free-to-download mobile application for women that creates an instant community safety network. At its core, the app connects women in distress with **verified SafeWalk Volunteers** registered within their local area. The app requires zero internet data for the panic alert — it uses SMS as a fallback.

Key Features

Feature	Description
Panic Button (1-tap SOS)	One tap sends a GPS location + distress alert to all verified volunteers within 500m AND to 3 pre-set emergency contacts simultaneously.
Live GPS Tracking	Contacts and responding volunteers can see the user's real-time location on a map.
SafeWalk Volunteer Badge	Volunteers undergo Aadhaar-based verification and wear a QR-coded identity badge at safe zones near stations.
Route Safety Rating	Women can rate the safety of streets/routes; unsafe areas are flagged on the in-app map.
Discreet Alert Mode	Phone appears to be playing music but is silently tracking and alerting — for situations where using the phone openly is risky.
Helpline Integration	One-tap access to Nirbhaya Helpline (1091), Mumbai Police (100), and nearest women's police station.

User Journey — Step by Step

Step	Action	What Happens
1	**Download & Register**	User downloads SafeWalk, registers with mobile number and sets 3 emergency contacts.
2	**Volunteer Joins**	Local volunteers register, verify identity, and get activated in the area network.
3	**User Starts Walking**	The user opens the app and taps 'I am walking alone'. App activates background GPS tracking.
4	**Distress Occurs**	The user presses the large red SOS button (or shakes the phone 3 times for a hands-free trigger).
5	**Instant Alert Sent**	All volunteers within 500m receive a push notification + SMS with the user's exact location.
6	**Volunteer Responds**	Nearest volunteer confirms response in app. User sees the volunteer's name, photo and ETA.
7	**Situation Resolved**	User marks 'I am safe'. The incident is anonymously logged to improve area safety ratings.

6 Competitive Analysis

Before finalising the SafeWalk solution, it is important to analyse existing safety tools and apps available in the Indian market. This helps position SafeWalk correctly and identify the gaps that it uniquely addresses.

Feature	SafeWalk Mumbai	Himmat (Delhi Police)	Nirbhaya (Govt)	bSafe (Global)	Google Maps
Community Volunteers	✅ Yes (500m radius)	❌ No	❌ No	⚠️ Friends only	❌ No
SMS Fallback (No Data)	✅ Yes	❌ No	❌ No	❌ No	❌ No
Discreet Alert Mode	✅ Yes	❌ No	❌ No	⚠️ Partial	❌ No
Shake-to-SOS	✅ Yes	✅ Yes	❌ No	✅ Yes	❌ No
Works for Low-Income Users	✅ Yes (SMS)	⚠️ Partial	⚠️ Partial	❌ No	❌ No
Hyperlocal to Mumbai	✅ Yes	❌ Delhi only	✅ Pan-India	❌ Global	✅ Global
Route Safety Ratings	✅ Yes	❌ No	❌ No	❌ No	⚠️ Partial

Competitive Advantage

SafeWalk Mumbai's primary competitive advantage lies in three areas that no existing app covers simultaneously:

- Community-powered real-time help within 500 metres — faster and more accountable than calling the police.
- SMS-based fallback that works without internet data — making it accessible to low-income users like Rekha who cannot afford data packs.
- Discreet alert mode for high-risk situations where drawing attention to the phone could itself escalate danger.

Government apps like Himmat (Delhi Police) and Nirbhaya are reactive and police-centric. International apps like bSafe rely on the user's personal network (friends, family) and have no community volunteer layer. SafeWalk is uniquely positioned at the intersection of community action, hyperlocal Mumbai knowledge, and digital inclusivity for lower-income users.

7 Prototype / Wireframe Description

As a next step in the Design Thinking process, I conceptualised a low-fidelity wireframe for the SafeWalk Mumbai app. While a clickable prototype would be built using tools like Figma or Marvel App, the following describes the key screens and interaction flows of the envisioned app.

Screen 1 — Onboarding & Registration

The first screen greets users with a simple, reassuring purple and white interface. It collects: mobile number (for SMS fallback), name, up to 3 emergency contacts with their numbers, and preferred language (English, Hindi, Marathi). Aadhaar verification is optional for regular users but mandatory for volunteers. The onboarding is designed to be completed in under 2 minutes.

Screen 2 — Home / Walking Mode Dashboard

The main screen shows a full-screen map of the user's current location, overlaid with colour-coded route safety ratings (green = safe, yellow = caution, red = reported unsafe). A large red SOS button occupies the bottom 30% of the screen — impossible to miss. A smaller 'I am walking alone' toggle activates background GPS logging. A discreet music-player icon in the top corner activates Discreet Mode.

Screen 3 — SOS Triggered State

When the SOS button is pressed (or the phone is shaken 3 times), the screen changes to a pulsing red state with the message 'Help is on the way.' The app simultaneously: sends a push notification + SMS to all volunteers within 500m, sends the GPS location + a distress message to the 3 pre-set emergency contacts, and begins a 60-second loop audio recording for evidence. The user can see a live list of responding volunteers with their distance and ETA.

Screen 4 — Volunteer App Interface

Volunteers see a separate but integrated version of the app. When a nearby SOS is triggered, they receive a loud alert with the user's location on a map. They can tap 'I am responding' to confirm. Their profile photo, name, and phone number are then shared with the user for transparency and accountability. After resolution, both parties rate the interaction.

Screen 5 — Route Rating & Community Map

After completing a walk, users are prompted to rate their route with a simple 1–5 star system and can optionally add a tag (e.g., 'no street lights', 'drunk men near bus stop', 'safe CCTV area'). These ratings are aggregated anonymously and displayed on the community map, helping future users plan safer routes. BMC-reportable dark lanes can be flagged with one tap.

> Design Principle: Every screen is designed for one-hand use, minimum text (for low-literacy users), and high contrast for night-time visibility. The app supports English, Hindi, and Marathi.

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8 Risk Analysis & Mitigation

Every new product faces risks. SafeWalk Mumbai is no exception. Below is an honest assessment of key risks, their likelihood, potential impact, and how the product team would mitigate them.

Risk	Likelihood	Impact	Mitigation Strategy
Volunteer reliability — volunteers may not respond in time	High	High	Implement a 'response rate' score visible on volunteer profiles. Volunteers with <70% response rate are temporarily deactivated. Partner with NGOs and college NSS units for a committed volunteer base.
False SOS triggers — accidental activations flood volunteers	Medium	Medium	Require a 3-second press (not single tap) for SOS. Add a 10-second cancel window. Track false trigger patterns and educate users in onboarding.
Data privacy — GPS data misuse or breach	Medium	High	All location data is encrypted end-to-end. No data is stored after the session ends. Incident logs are anonymised. Comply with India's DPDP Act 2023.
Low volunteer adoption — not enough volunteers near users	High	High	Launch first in high-density areas (Andheri, Bandra, Dadar). Partner with colleges for NSS credit for volunteering. Run awareness campaigns at local train stations.
Misuse by bad actors — abusers masquerading as volunteers	Medium	Very High	Mandatory Aadhaar + selfie verification for all volunteers. Volunteers are visible to users only after SOS; they cannot proactively contact users. Background check integration with DigiLocker.
Low smartphone literacy among low-income users	Medium	Medium	Maximise simplicity: only one main button on the home screen. Offer a feature phone USSD-based fallback (*123# shortcode). Partner with NGOs for onboarding workshops in Dharavi and other dense localities.
Regulatory hurdles — RBI/telecom licensing for SMS services	Low	Medium	Partner with established SMS gateway providers (MSG91, Textlocal) who hold required DLT registrations. Engage legal counsel early for compliance.

9 Basic Business Model

Component	Details
Target Users	Women aged 16–45 in Mumbai (Phase 1); expand to Pune, Bengaluru, Delhi (Phase 2). Secondary users: safety-conscious parents, corporate HR departments.
Revenue Model	Freemium: Basic app free for all users. B2B subscriptions for corporates (employee safety dashboard). Government/CSR grants from NGOs and civic bodies. Premium subscription (Rs 49/month) for advanced features.
Pricing	Individual: Free (basic) / Rs 49/month (premium). Corporate: Rs 5,000/month per 100 employees.
Cost Structure	App development: Rs 3–5 lakhs (one-time). Server & SMS costs: Rs 30,000/month. Volunteer onboarding: Rs 10,000/month. Marketing: Rs 20,000/month. Total monthly operating cost: Rs 60,000–80,000.

Customer Acquisition Cost (CAC)

Definition: CAC = Total Marketing Spend ÷ Number of New Users Acquired**Example for SafeWalk Mumbai:**Marketing spend in Month 1 = Rs 20,000 (college campus drives, Instagram ads)New users acquired = 400 women**CAC = Rs 20,000 ÷ 400 = Rs 50 per user**This is a very low CAC, achievable through word-of-mouth referrals and college partnerships (zero-cost). Women naturally share safety tools with friends.

Contribution Margin

Definition: Contribution Margin = Selling Price − Variable Cost per unit**Premium Subscription Example:**Selling Price = Rs 49/month | Variable Cost = Rs 8/month (SMS + server)**Contribution Margin = Rs 49 − Rs 8 = Rs 41 per user per monthCorporate Plan Example:**Selling Price = Rs 5,000/month | Variable Cost = Rs 800/month**Contribution Margin = Rs 4,200 per corporate client per month**

Can This Be Profitable?

Yes — with **20 corporate clients** at Rs 5,000/month, monthly revenue = Rs 1,00,000, which covers all operating costs (Rs 60,000–80,000) and leaves a positive margin. The freemium individual model drives user growth and social credibility. Long-term, government safety contracts and CSR funding from large corporations (Tata, Reliance) can provide additional capital. This idea is also eligible for grants under the **Nirbhaya Fund** and **StartupIndia scheme.**

10 Implementation Roadmap

SafeWalk Mumbai will be launched in three phases over 18 months. The roadmap is designed to validate the concept with a small community before scaling city-wide and then nationally.

Phase	Timeline	Key Activities
Phase 1 — Build & Pilot	Months 1–4	Finalise app features and wireframes. Hire a 2-person dev team (or use an agency). Build MVP (Minimum Viable Product) with Panic Button, GPS, SMS fallback, and volunteer registration. Recruit 50 volunteers in Andheri and Bandra. Soft-launch with 200 beta users (college students via NSS units). Gather feedback and fix bugs.
Phase 2 — Refine & Grow	Months 5–10	Incorporate beta feedback: add route safety ratings, discreet mode, and helpline integration. Launch at 10 Mumbai college campuses. Run social media campaigns. Approach 5 corporate clients (IT parks, hospitals) for B2B pilots. Apply for Nirbhaya Fund grant and StartupIndia registration. Expand to 1,000+ users.
Phase 3 — Scale & Sustain	Months 11–18	Expand to all major Mumbai localities (Dharavi, Kurla, Thane). Partner with Mumbai Police and BMC for official endorsement. Launch Marathi and Hindi versions. Onboard 20+ corporate clients. Explore partnerships with Uber/Ola for auto/cab safety integration. Begin feasibility study for Pune and Bengaluru expansion. Target: 50,000 users, 2,000 volunteers.

Key Milestones

Month	Milestone
Month 1	App wireframes complete; development begins
Month 3	MVP built and ready for beta testing
Month 4	50 volunteers onboarded; pilot launched in Andheri & Bandra
Month 6	500 users; first corporate client signed
Month 9	Nirbhaya Fund grant application submitted
Month 12	10,000 users; 500 volunteers; break-even on operating costs
Month 18	50,000 users; expansion to second city

11 Validation Exercise

After developing the SafeWalk concept, I presented it to the same three users and asked three key questions. Below are their responses:

User	Would you use this?	What you liked	What you'd improve
Priya (Student)	Yes, definitely! I would download it right away.	The shake-to-SOS feature is genius — I can do it while phone is in bag.	Add a feature to rate auto drivers specifically.
Fatima (Nurse)	100% yes. Even just live tracking would help my husband worry less.	Volunteer network idea is great. Discreet mode is very thoughtful.	Need to show volunteer response rate for accountability.
Rekha (Sweeper)	Yes, if it works on SMS without internet.	The SMS fallback alert is perfect for someone like me.	Make the app lighter and add a feature to report dark roads to BMC directly.

Validation Summary

What worked: All 3 users responded very positively. The SOS + volunteer network idea was universally liked. The discreet mode feature was highlighted as particularly thoughtful and practical.

What needs improvement: Users raised concerns about volunteer reliability — the system needs a rating/accountability mechanism. Chat features and driver ratings were also requested. These will be incorporated in Version 2 of the app.

12 Reflection and Learning

What I Learned

This Case Study fundamentally changed how I think about problem-solving. Before beginning, I assumed women's safety apps already existed and the problem was solved. But talking to real users revealed a deep truth: existing tools are reactive, not proactive, and they place the entire burden on the victim. The Design Thinking process forced me to listen first and design second — which led to a much more human-centred solution.

What Surprised Me

The most surprising finding was that all three women had already developed their own informal safety systems (constant WhatsApp check-ins, keeping someone on call, memorising safe routes). This showed me that the need is so urgent that people will create workarounds even without a proper solution. It also showed that women are incredibly resourceful — the right tool just needs to formalise and amplify what they already do naturally.

What I Would Improve

If I were to extend this project, I would:

- Conduct a wider survey with 20+ women across different income groups and areas of Mumbai.
- Build a low-fidelity prototype (wireframe) of the app and test actual usability with real users.
- Speak with Mumbai Police and NGOs like iCall and Majlis to understand legal and on-ground challenges.
- Explore partnerships with colleges and IT parks for a pilot launch.
- Develop a more rigorous financial model with real data on volunteer retention and corporate sales cycles.

Personal Reflection

Design Thinking taught me that empathy is not a soft skill — it is the most powerful engineering tool we have. A solution built from real user pain is always more effective than one built from assumptions.

I also learned that entrepreneurship is not just about making money; it is about identifying a real problem and having the courage to solve it, even if the market is not obvious at first. The three women I interviewed — Priya, Fatima, and Rekha — represent millions of women across Mumbai who deserve better. SafeWalk Mumbai is my attempt to give them a voice and a tool.

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